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ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Project Report
CSE 4206: Digital Logic Design Lab
Summer 24-25
Ship Cargo Loading System

Team Members

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1. Project Details

Field	Details
Project Title	Ship Cargo Loading System
Course	CSE 4206 - Digital Logic Design Lab
Semester	Summer 2024–2025
Institution	Islamic University of Technology (IUT)
Simulation Tool	Proteus 8 Professional

2. Overview

The Ship Cargo Loading System is a simulation-based digital logic circuit that models a port authority's single-dock cargo management infrastructure, implemented and fully verified in Proteus. Ships arrive at the dock carrying binary-encoded cargo weights (4-bit, 0–15 units per vessel). The system accumulates the total dock load in a synchronous register, continuously compares it against an operator-set ship capacity threshold, and drives a tri-LED status alarm to signal one of three states: under-capacity, overload, or a full-ship departure condition.

When the dock weight equals the ship's rated capacity - a full-ship departure event - two simultaneous automated actions are triggered: the ship departure counter (displayed on a 7-segment display) increments by one, and the dock register is cleared to zero, immediately readying the dock for the next vessel. Dock weight is decoded from binary into BCD and presented on a dual 7-segment display supporting values from 0 to 19 units.

The system is structured as a five-phase combinational and sequential pipeline using standard 74LS-series TTL ICs throughout, covering all seven required component categories: inputs, adders, encoders/decoders, flip-flops, registers, counters, and output devices.

3. Implemented Features

The system is organized into five functional subsystems. Each subsystem operates continuously whenever the circuit is powered; there is no explicit mode switching.

3.1 Cargo Input Validation & Pulse Generator

The operator encodes incoming ship cargo weight using four LOGICSTATE switches (SW0 = 1, SW1 = 2, SW2 = 4, SW3 = 8), producing a 4-bit binary value (0–15 units). An OR-tree gating network built from three 74LS32 quad OR gate sub-gates (U1:A, U1:B, U1:C) combines all four switch lines. If all inputs are LOW (zero-weight - empty ship arrival), the

OR-tree output remains LOW and no pulse is generated, rejecting empty arrivals without a dedicated encoder IC.

When at least one switch is HIGH, a 74LS08 AND gate (U2:A) gated with an active-HIGH enable signal produces a validated arrival signal. A 74LS04 NOT gate (U8:A) inverts this to an active-LOW pulse on the Parallel Load pin (PL, Pin 11) of the dock register, capturing the current adder output into the accumulator.

3.2 Synchronous Dock Weight Accumulator & Feedback Loop

A 74LS193 4-bit synchronous up/down counter (U6) serves as the dock weight storage register. A 74LS283 4-bit adder (U7) continuously sums the incoming cargo weight (SW0–SW3) with the current register state (Q0–Q3 of U6), producing the next accumulated total (S0–S3). This forms a closed feedback loop: each validated ship arrival loads the new sum back into U6.

A 10 k Ω pull-up resistor (R1) connects VCC to the active-LOW Parallel Load pin (Pin 11) of U6, holding PL HIGH by default. This eliminates spurious register loads during Proteus SPICE transient analysis by ensuring the register only updates on an explicit active-LOW pulse from the input validation subsystem. Register outputs (Q0–Q3) simultaneously feed the BCD display subsystem, the magnitude comparator, and the feedback adder.

3.3 BCD Range Corrector & Dual 7-Segment Dock Weight Display

The raw 4-bit binary sum from the accumulator may exceed 9, requiring BCD correction for valid decimal display. A combinational detection network using 74LS08 AND gates (U2:B, U2:C) and a 74LS32 OR gate (U1:D) evaluates the condition $(Q3 \cdot Q2) \text{ OR } (Q3 \cdot Q1)$. When this expression is HIGH, the raw sum is ≥ 10 and correction is required.

A 74LS283 BCD correction adder (U5) adds a +6 offset (0110) to the raw accumulator outputs when the correction flag is asserted. Two 74LS47 BCD-to-7-segment decoders drive Common Anode displays: U3 drives the Tens digit (outputting 0 or 1 based on whether the sum exceeds 9), and U4 drives the corrected Units digit. Control pins LT (Pin 3), BI/RBO (Pin 4), and RBI (Pin 5) on both decoders are tied HIGH (VCC) to keep displays permanently active. The system accurately represents dock weights from 00 to 19.

3.4 Magnitude Comparator & Tri-LED Status Alarm

A 74LS85 4-bit magnitude comparator (U9) compares the live dock register outputs (A0–A3, from Q0–Q3 of U6) against operator-set capacity threshold switches (B0–B3). Cascade inputs are configured as follows: Pin 3 (A=B in) is tied HIGH (VCC); Pins 2 and 4 (A<B in and A>B in) are tied LOW (GND), configuring U9 as a standalone comparator with no upstream cascade.

Three status LEDs, each driven through a 220 Ω current-limiting resistor, indicate system state:

- Yellow LED D1 - driven by QA<B (Pin 7): dock weight is below rated capacity; system is accepting cargo normally.
- Red LED D2 - driven by QA>B (Pin 5): dock weight exceeds rated capacity; overload alarm condition.

- Green LED D3 - driven by QA=B (Pin 6): dock weight exactly matches rated capacity; full-ship departure event triggered.

3.5 Ship Departure Counter & Automated Dock Reset

The QA=B output (Pin 6 of U9) fans out simultaneously to three destinations, forming the automated departure and reset pipeline:

- Clock input CKA (Pin 14) of U10 - increments the ship departure counter.
- Anode of Green LED D3 - illuminates the full-ship departure notification LED.
- Master Reset MR (Pin 14) of U6 - clears the dock weight register to zero, immediately readying the dock for the next vessel.

The 74LS90 decade counter (U10) operates in BCD mode: output Q0 is wired to clock input CKB (Pin 1); reset pins R0(1), R0(2) and R9(1), R9(2) are all grounded, disabling manual reset and 9's complement modes. U10 counts from 0 to 9, tracking total ship departures per session. A 74LS47 BCD-to-7-segment decoder (U11) drives a single Common Anode 7-segment display showing the running departure total.

3.6 Component Inventory

Designator	IC / Component	Function
U1	74LS32 - Quad OR	OR-tree gating (U1:A/B/C) + BCD correction detection (U1:D)
U2	74LS08 - Quad AND	Cargo pulse gating (U2:A) + BCD detection (U2:B/C)
U3	74LS47 - BCD-to-7-Seg	Tens digit decoder - dock weight display
U4	74LS47 - BCD-to-7-Seg	Units digit decoder - dock weight display
U5	74LS283 - 4-bit Adder	BCD +6 offset corrector
U6	74LS193 - 4-bit Up/Down Counter	Dock weight accumulator / storage register
U7	74LS283 - 4-bit Adder	Cargo accumulation feedback adder
U8	74LS04 - Hex Inverter	Active-HIGH to active-LOW PL pulse inversion (U8:A)
U9	74LS85 - 4-bit Comparator	Dock weight vs. capacity threshold comparison
U10	74LS90 - Decade Counter	Ship departure counter (BCD mode, 0–9)

Designator	IC / Component	Function
U11	74LS47 - BCD-to-7-Seg	Ship departure count display decoder
R1	10 k Ω resistor	Pull-up on U6 PL (Pin 11) to VCC
D1	Yellow LED + 220 Ω	Under-capacity status indicator
D2	Red LED + 220 Ω	Overload alarm indicator
D3	Green LED + 220 Ω	Full-ship / departure triggered indicator
-	Common Anode 7-Seg $\times$ 3	Dock weight tens, dock weight units, ship departure count

4. Screenshots & Explanations

The following screenshots capture the four key functional subsystems of the circuit. Each figure corresponds to an independently wired module with clearly defined input/output signal interfaces to adjacent subsystems.

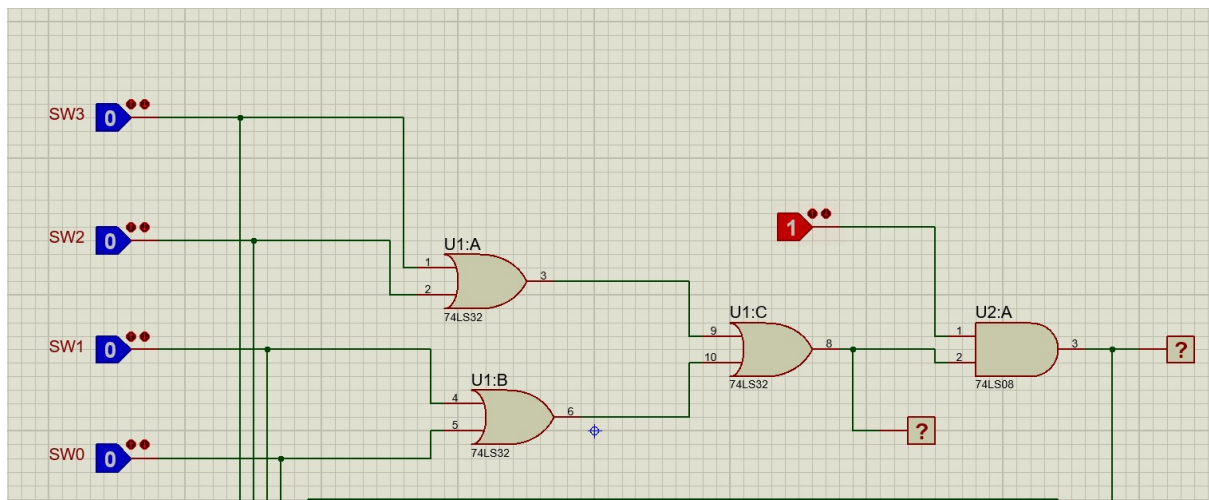


Figure 1: Cargo Input Validation & Pulse Generator Subsystem

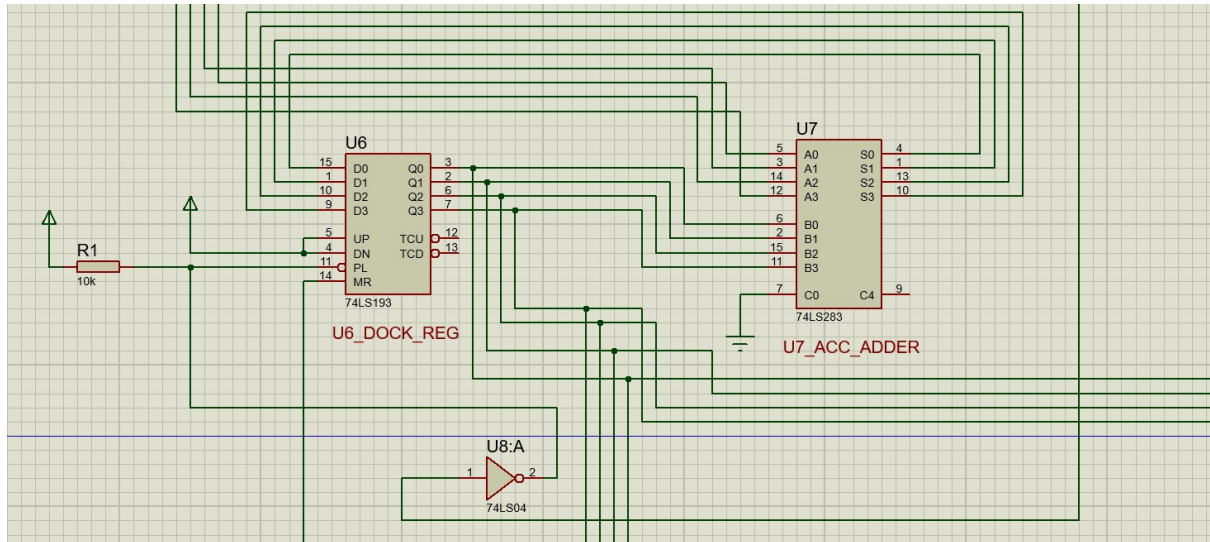


Figure 2: Synchronous Dock Weight Accumulator & Feedback Adder Loop

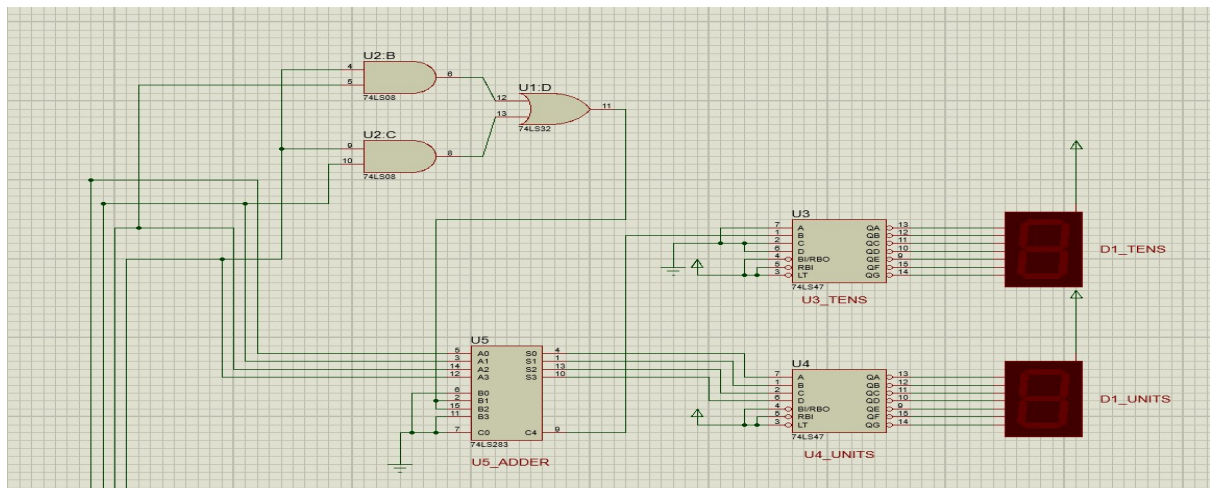


Figure 3: BCD Range Corrector Network & Dual 7-Segment Display Drivers (0-19 Range)

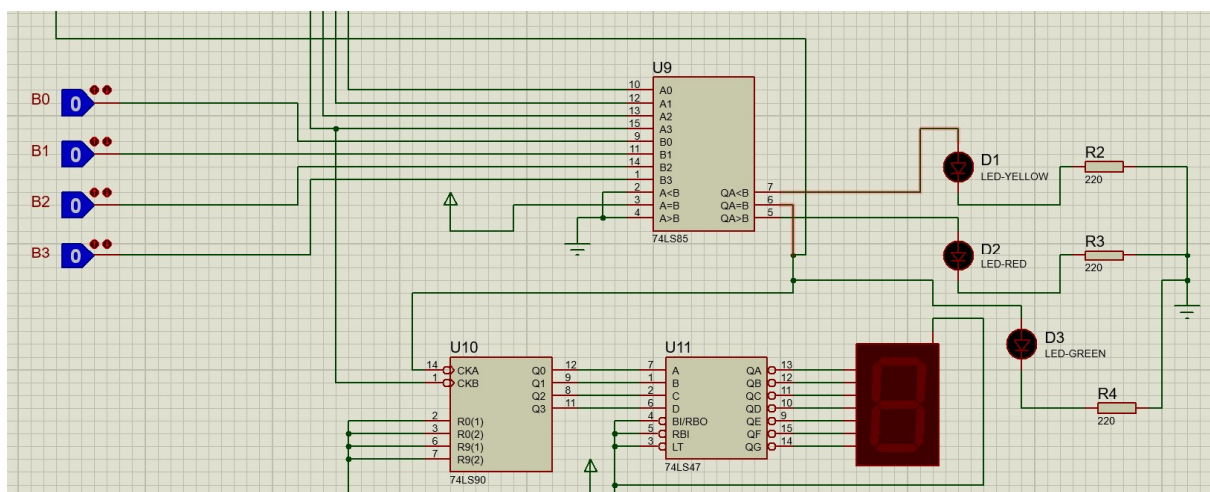


Figure 4: Capacity Magnitude Comparator, Status Indicators & Automated Ship Departure Counter

5. Changes from the Proposal

The finalized circuit differs substantially from the initial project proposal. The changes below record each design evolution, its technical motivation, and the engineering trade-off accepted.

5.1 Unimplemented Features

5.1.1 Multi-State FSM Engine (Removed)

The original proposal included a 5-state sequential FSM (IDLE → LOADING → OVERFLOW → STORM → DRAINING) implemented using JK or D flip-flops with combinational next-state and output logic.

What replaced it: Direct, combinational comparator outputs from the 74LS85 (U9). System behavior is fully determined at any moment by the instantaneous relationship between the dock register value and the capacity threshold - no state variables are required.

Reason for removal:

- The FSM introduced multiple flip-flop propagation delays in series with the comparator path. In Proteus transient simulation, these cascaded delays produced glitchy intermediate states at clock edges, causing the FSM to settle in undefined transitions.
- State-machine lockup - where the FSM could enter a state with no valid exit under certain input sequences - required additional combinational guard logic, increasing circuit area without functional benefit.
- The 74LS85's three output pins ($QA < B$, $QA > B$, $QA = B$) already encode all three operationally meaningful conditions as mutually exclusive combinational signals, directly usable without sequential wrapping.
- Eliminating the FSM removed approximately 6–8 flip-flop ICs and all state-encoding overhead, while making circuit behavior fully transparent and verifiable in a single simulation pass.

5.1.2 Storm Drain / Bilge Purge Subsystem (Removed)

The original proposal included a storm-mode subsystem activated by a toggle switch, transitioning the FSM into a DRAINING state. A down-counter would decrement both dock load registers simultaneously at a fixed rate; a Ring/Johnson counter would drive a drain animation LED sequence. The system would auto-return to IDLE when both docks reached zero.

What replaced it: Automated dock reset via the comparator's $QA = B$ output directly asserting the MR (Master Reset) pin of U6, eliminating any explicit drain mode.

Reason for removal:

- The drain subsystem required a continuous feedback loop between the down-counter and the register. In Proteus SPICE, this loop created a matrix convergence problem - the Newton-Raphson solver could not converge on a stable operating point at each timestep, manifesting as simulation reload loops and SPICE matrix singularity warnings.
- The Ring/Johnson counter animation had no functional role in the cargo logic and added three or more ICs purely for aesthetic output.

- The automated $QA=B \rightarrow MR$ path achieves the same dock-clearing outcome with zero additional ICs, no feedback instability, and deterministic single-cycle reset.

5.2 Modified Features

5.2.1 Elimination of Manual Reset Bus Contention

The original proposal included dedicated LOGICSTATE switches connected to the Master Reset (MR, Pin 14) of the 74LS193 dock register, allowing the operator to manually clear the dock weight at any time.

What replaced it: MR is driven exclusively by the $QA=B$ output of U9 (Pin 6), with no operator switch on the line.

Reason for change:

- Connecting both a LOGICSTATE switch and the $QA=B$ comparator output to the MR pin created a wired-OR bus conflict, producing bus conflict warnings in Proteus and unpredictable reset behavior.
- Resolving the conflict would have required an OR gate on the MR line, introducing an additional IC, an extra gate delay on the reset path, and priority arbitration logic.
- The $QA=B$ -driven auto-reset is semantically correct: the dock clears precisely when a ship is fully loaded, which is the only operationally valid time for a dock reset. A manual override at arbitrary times could corrupt the accumulation state and produce incorrect departure counts.

5.3 Additional Features

5.3.1 Pull-Up Resistor Stabilization on Register PL Pin

A 10 k Ω pull-up resistor (R1) was added between VCC and Pin 11 (PL) of the 74LS193 dock register - not present in the original proposal.

Reason for addition:

- In Proteus, undriven active-LOW control pins on synchronous ICs do not simulate as simple HIGH. During transient analysis initialization, a floating PL pin produces an indeterminate voltage that the SPICE solver can interpret as LOW, causing the register to spuriously parallel-load before any cargo input is applied.
- The 10 k Ω value is chosen so that the active-LOW pulse driver (U8:A, 74LS04, rated to sink up to 8 mA) can cleanly override it - at $VCC = 5\text{ V}$, the pull-up sources only 0.5 mA against the inverter's sink capacity - ensuring the PL pin reliably reaches logic LOW during a valid pulse without contention.

6. Individual Contributions

Team responsibilities map directly onto the five-phase pipeline. Each member owns a self-contained set of subsystems with explicit signal interfaces to adjacent modules.

MD. Sadman Saif Zarif (Member 1) - Input & Storage Lead

Assigned Modules	Input Validation (Subsystem 1) & Dock Accumulator Loop (Subsystem 2)
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Core Hardware Owned

- Switches: SW0 (1), SW1 (2), SW2 (4), SW3 (8)
- Logic Gates: 74LS32 OR-tree (U1:A, U1:B, U1:C), 74LS08 gating (U2:A), 74LS04 inverter (U8:A)
- Storage & Math: 74LS193 4-bit synchronous register (U6), 74LS283 4-bit adder (U7), 10 kΩ pull-up resistor (R1)

Technical Responsibilities

1. Build and verify the OR-tree logic to detect any incoming cargo weight trigger.
2. Implement the active-LOW load pulse generator feeding Pin 11 (PL) of the 74LS193 register.
3. Stabilize the register using the 10 kΩ pull-up resistor on Pin 11 to eliminate Proteus SPICE matrix convergence errors.
4. Wire the closed-loop feedback path between adder outputs (S0–S3) and register inputs (D0–D3).

Lab Report Sections Owned

- Section 3.1: Input Validation Circuit Schematic & Pulse Generator Analysis
- Section 3.3: Synchronous Register Dynamics & Adder Feedback Mechanics
- Section 5.1: SPICE Simulation Stability & Pull-Up Resistor Troubleshooting

Ayesha Chowdhury Aronti (Member 2) - Visualization Lead

Assigned Modules	BCD Range Corrector & Dual Dock Weight Displays (Subsystem 3)
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Core Hardware Owned

- Logic Gates: 74LS08 AND gates (U2:B, U2:C), 74LS32 OR gate (U1:D)
- BCD Conversion: 74LS283 4-bit binary adder (U5)
- Displays: Dual 74LS47 BCD-to-7-segment decoders (U3 Tens, U4 Units), Dual Common Anode 7-segment displays

Technical Responsibilities

1. Design the combinational detection network to flag binary accumulator sums ≥ 10 [expression: $(Q3 \cdot Q2) \text{ OR } (Q3 \cdot Q1)$].
2. Wire the 74LS283 BCD correction adder (U5) to inject a +6 offset (0110) when sums exceed 9.
3. Route corrected BCD lines to the Units display decoder (U4) and raw overflow bits to the Tens display decoder (U3).
4. Ensure proper common anode power routing and verify control pins (LT, BI/RBO, RBI) are tied HIGH to prevent display dimming or glitching.

Lab Report Sections Owned

- Section 3.2: Binary-to-BCD Conversion Theory & Range Extension (0–19)
- Section 4.1: Display Decoding & Segment Mapping Diagnostics

Sanjid Ahsan (Member 3) - Automation Lead

Assigned Modules	Magnitude Comparator (Subsystem 4) & Ship Counter & Auto-Reset (Subsystem 5)
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Core Hardware Owned

- Comparator: 74LS85 4-bit magnitude comparator (U9), capacity limit switches B0–B3
- Status Indicators: Yellow LED D1 (Under Capacity), Red LED D2 (Overload), Green LED D3 (Full Capacity), 220 Ω current-limiting resistors (R2–R4)
- Counter & Reset: 74LS90 decade counter (U10), 74LS47 decoder (U11), single Common Anode 7-segment display, automated reset feedback trace

Technical Responsibilities

1. Wire live register outputs (A0–A3) and capacity switches (B0–B3) into the 74LS85 comparator (U9), and configure cascade pins (Pin 3 = VCC, Pins 2 & 4 = GND).
2. Implement current-limited LED status outputs for $A < B$ (Yellow) and $A > B$ (Red) conditions.
3. Configure the 74LS90 decade counter (U10) in 0–9 BCD decade mode with Q0 tied to CKB and all reset pins grounded.
4. Establish the automated parallel reset line linking Pin 6 (QA=B) of U9 simultaneously to the Green LED anode, U10 clock pin 14, and U6 master reset pin 14 (MR).

Lab Report Sections Owned

- Section 3.4: Comparator Logic & Status Alarm Subsystem
- Section 3.5: Automated Departure Counter & Zero-Delay Dock Reset Loop
- Section 6.1: End-to-End System Integration & State Transition Analysis

Jointly Completed

The following components were designed or verified collaboratively by all three members:

- End-to-end signal interface verification across all five subsystem boundaries.
- Cross-subsystem wire routing in Proteus (inter-module nets: PL pulse, Q0–Q3 fan-out, MR feedback trace).
- Full-circuit functional walkthrough and demo script validation against the six-step test sequence.

7. Video Link

Video Title: CSE 4206 Project - Ship Cargo Loading System

Platform: YouTube

Link: https://youtu.be/494xMRGCugs?si=M_apV9tbnf73au61